

Lecture Notes II

Course: Functional Analysis

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Topic: Measurable Functions

Proof 3.8. Let $(X, \mathcal{M})$, $(Y, \mathcal{N})$, $(Z, \mathcal{O})$, then if $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ are measurable we have

$$f^{-1}(\mathcal{N}) \subseteq \mathcal{M} \quad \text{and} \quad g^{-1}(\mathcal{O}) \subseteq \mathcal{N},$$

moreover, notice how

$$(g \circ f)^{-1}(\mathcal{O}) = (f^{-1} \circ g^{-1})(\mathcal{O}) = f^{-1}(g^{-1}(\mathcal{O})) \subseteq f^{-1}(\mathcal{N}) \subseteq \mathcal{M}.$$

Proof 3.9. Suppose $f : X \rightarrow Y$ is continuous, then

$$f^{-1}(\mathcal{T}_Y) \subseteq \mathcal{T}_X.$$

Equip X and Y with their respective Borel σ -algebra. Let

$$\mathcal{N} = \{B \subseteq Y : f^{-1}(B) \in \mathcal{B}_X\}.$$

This is a σ -algebra on Y since

$$f^{-1}(\emptyset) = \emptyset, \quad f^{-1}(B^c) = (f^{-1}(B))^c, \quad f^{-1}\left(\bigcup_{n \in \mathbb{N}} B_n\right) = \bigcup_{n \in \mathbb{N}} f^{-1}(B_n).$$

Now, if $U \in \mathcal{T}_Y$, then by continuity of f we have $f^{-1}(U) \in \mathcal{T}_X$ and thus $f^{-1}(U) \in \mathcal{B}_X$ (because $\mathcal{B}_X \supset \mathcal{T}_X$) and hence $U \in \mathcal{N}$, so that $\mathcal{T}_Y \subseteq \mathcal{N}$. Furthermore, since $\mathcal{B}_Y = \sigma(\mathcal{T}_Y)$, then we must have $\mathcal{B}_Y \subseteq \mathcal{N}$ and therefore $f^{-1}(\mathcal{B}_Y) \subseteq f^{-1}(\mathcal{N})$, moreover, by definition of $\mathcal{N}$ we have $f^{-1}(\mathcal{N}) \subseteq \mathcal{B}_X$. We conclude that $f^{-1}(\mathcal{B}_Y) \subseteq \mathcal{B}_X$. $\square$

Proof 3.10. Let $(X, \mathcal{M})$ and $(Y, \mathcal{N})$ be two measurable spaces, and consider a function $f : X \rightarrow Y$. Suppose there is a subclass $\mathcal{C}$ of $\mathcal{N}$ such that $\sigma(\mathcal{C}) = \mathcal{N}$ and $f^{-1}(\mathcal{C}) \subseteq \mathcal{M}$. Then define

$$\mathcal{A} = \{A \subseteq Y : f^{-1}(A) \in \mathcal{M}\}.$$

It is clear (from the previous theorem) that $\mathcal{A}$ is a σ -algebra, then take $C \in \mathcal{C}$ so that the assumption implies $f^{-1}(C) \in \mathcal{M}$ and thus $C \in \mathcal{A}$, this shows that $\mathcal{C} \subseteq \mathcal{A}$, and since $\sigma(\mathcal{C})$ is the smallest σ -algebra containing $\mathcal{C}$, we get $\mathcal{C} \subseteq \sigma(\mathcal{C}) \subseteq \mathcal{A}$, moreover, by hypothesis, $\sigma(\mathcal{C}) = \mathcal{N}$ so that $\mathcal{N} \subseteq \mathcal{A}$ and thus $f^{-1}(\mathcal{N}) \subseteq f^{-1}(\mathcal{A})$. It is readily shown that $f^{-1}(\mathcal{A}) \subseteq \mathcal{M}$ so that using the previous result we get

$$f^{-1}(\mathcal{N}) \subseteq \mathcal{M}.$$

$\square$

Proof of $\mathcal{B}(\mathbb{R}^2) = \mathcal{B}(\mathbb{R}) \otimes \mathcal{B}(\mathbb{R})$. We know that the set

$$\mathcal{B} = \{(a, b) \times (c, d) : a, b, c, d \in \mathbb{Q}\}$$

forms a countable basis for the Euclidean topology $\mathcal{T}_{\mathbb{R}^2}$ of $\mathbb{R}^2$. Then we have that any element U of $\mathcal{T}_{\mathbb{R}^2}$ can be written as

$$U = \bigcup_{B \in \mathcal{B}} B,$$

which is a countable union since $\mathcal{B} \subseteq \mathbb{Q}^4$ and $\mathbb{Q}^4$ is countable. Since (a, b) and (c, d) are both in $\mathcal{B}(\mathbb{R})$ we conclude that

$$(a, b) \times (c, d) \in \mathcal{B}(\mathbb{R}) \otimes \mathcal{B}(\mathbb{R}),$$

but $\mathcal{B}(\mathbb{R}) \otimes \mathcal{B}(\mathbb{R})$ is a σ -algebra, so it is closed under countable unions and thus U is in $\mathcal{B}(\mathbb{R}) \otimes \mathcal{B}(\mathbb{R})$

$$\mathcal{B}(\mathbb{R}^2) \subseteq \mathcal{B}(\mathbb{R}) \otimes \mathcal{B}(\mathbb{R}).$$

Conversely, if $A, B \in \mathcal{B}(\mathbb{R})$, then since the coordinate projections $\pi_1, \pi_2 : \mathbb{R}^2 \rightarrow \mathbb{R}$ are continuous and hence Borel measurable, meaning that

$$\pi_1^{-1}(\mathcal{B}(\mathbb{R})) \subseteq \mathcal{B}(\mathbb{R}^2) \quad \text{and} \quad \pi_2^{-1}(\mathcal{B}(\mathbb{R})) \subseteq \mathcal{B}(\mathbb{R}^2)$$

so that we have (σ -algebras are closed under finite intersections)

$$A \times B = \pi_1^{-1}(A) \cap \pi_2^{-1}(B) \in \mathcal{B}(\mathbb{R}^2),$$

and then

$$\mathcal{C} = \{A \times B : A, B \in \mathcal{B}(\mathbb{R})\} \subseteq \mathcal{B}(\mathbb{R}^2),$$

now, since $\sigma(\mathcal{C})$ is the smallest σ -algebra containing $\mathcal{C}$ and $\mathcal{B}(\mathbb{R}^2)$ is some σ -algebra containing $\mathcal{C}$, we must have $\sigma(\mathcal{C}) \subseteq \mathcal{B}(\mathbb{R}^2)$ and notice that $\sigma(\mathcal{C})$ is just the definition of $\mathcal{B}(\mathbb{R}) \otimes \mathcal{B}(\mathbb{R})$ so that $\mathcal{B}(\mathbb{R}) \otimes \mathcal{B}(\mathbb{R}) \subseteq \mathcal{B}(\mathbb{R}^2)$

$$\mathcal{B}(\mathbb{R}) \otimes \mathcal{B}(\mathbb{R}) \subseteq \mathcal{B}(\mathbb{R}^2).$$

We therefore conclude that $\mathcal{B}(\mathbb{R}^2) = \mathcal{B}(\mathbb{R}) \otimes \mathcal{B}(\mathbb{R})$. □

Corollary. $\mathcal{B}(\mathbb{R}^n) = \mathcal{B}(\mathbb{R})^{\otimes n}$.